• CHEERIO AI • PRODUCT STRATEGY MEMORANDUM

Assignment A: Scaling Customer Support for Non-Technical Users

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Executive Summary: Cheerio supports 5,000 active Indian SMB merchants generating ~17,500 support queries/month. The core problem is not an FAQ deficiency; it is technical onboarding friction across Meta Cloud API compliance, phone binding, and CSV formatting. Instead of a conversational chatbot that requires users to explain technical errors, we designed **Cheerio Co-Pilot**: an embedded, state-aware widget providing 1-click deterministic remediations. This strategy reduces support cost per ticket by 62.4% (from ₹40.00 to ₹15.06), avoids 13 support hires as volume scales, generates ₹52.38L in net annual operational savings, and delivers a 50.4× net ROI on monthly AI spend by curbing merchant churn.

1.1 THE CORE SCALING BOTTLENECK FOR NON-TECHNICAL MERCHANTS

Cheerio empowers non-technical business owners (retail jewellers, D2C brands, clinics, coaching centers) to run WhatsApp marketing campaigns. However, these users struggle with strict developer platforms: Meta Business Manager, Cloud API error payloads, 2FA coexistence traps, and strict CSV schemas. When a template is rejected or a phone number fails to bind, merchants cannot diagnose the technical root cause. They submit repetitive, high-touch support tickets ("WhatsApp is broken", "My blast won't send"), overwhelming Cheerio's human support operations.

1.2 ONBOARDING BOTTLENECKS & FAILURE TAXONOMY (5,000 MERCHANT SCALE)

ONBOARDING BOTTLENECK	EST. VOLUME	UNDERLYING TECHNICAL CAUSE	MERCHANT EXPERIENCE & FAILURE IMPACT
Template Rejections	38% (6,650/mo)	Meta rejection `INCORRECT_CATEGORY` (promotional discount copy submitted under Utility category).	Template rejected by Meta; campaign delayed; merchant frustrated by perceived tool bug.
WABA Phone Binding	27% (4,725/mo)	Meta Error `133010` (Account not registered / phone locked to consumer WhatsApp 2FA).	Merchant stuck on phone connection; cannot broadcast; high churn risk during Day 1–3.
CSV Contact Uploads	19% (3,325/mo)	Missing +91 international dialing prefixes; illegal spaces, hyphens, and empty rows.	Broadcast campaign aborts on upload; merchant unable to fix 1,000+ Excel rows manually.
Meta Verification	11% (1,925/mo)	Two-factor auth lockouts, missing GST business proofs, or rejected display names.	Blocked by Meta Business Suite; merchant blames Cheerio for external Meta platform rules.
Webhook & Cloud Sync	5% (875/mo)	Shopify / WooCommerce sync drops, expired access tokens, catalog synchronization lags.	Abandoned cart triggers do not fire; attribution tracking breaks silently.

[Candidate estimate: 38/27/19/11/5 distribution based on typical WhatsApp onboarding bottlenecks; 3.5 queries/merchant/month across 5,000 active businesses].

1.3 SOLUTION SPACE EXPLORATION: 8 ALTERNATIVES EVALUATED & SCORED

To avoid premature convergence on a standard chatbot, 8 distinct operational interventions were scored across Deflection Potential, Engineering Effort, User Friction, and Operational Risk:

SOLUTION ALTERNATIVE	DEFLECTION	DEV EFFORT	USER FRICTION	RISK	STRATEGIC VERDICT
1. In-App Context Co-Pilot	**High (65%)**	**Medium (1 mo)**	**Near-Zero**	**Low**	**Selected: Primary Core Product**
2. Proactive Input Linter	Medium (35%)	Low (2 wks)	**Near-Zero**	Very Low	**Hybridized into Co-Pilot frontend**
3. Guided Onboarding Checklist	Low (15%)	Low (1 wk)	Low	Low	Good orientation, but cannot fix broken inputs
4. WhatsApp Support Bot	Medium (40%)	Medium (3 wks)	Medium	High	Forces copy-pasting error payloads across apps
5. Video & Community Library	Very Low (8%)	Low (1 wk)	High	Low	Passive; busy SMB owners do not watch tutorials
6. Done-For-You (DFY) Service	High (80%)	None	**Zero**	**Severe (Cost)**	Unscalable; unit economics collapse on margins
7. Agent-Assist Internal Copilot	Zero (0%)	Medium (3 wks)	None	Low	Improves agent handle time, but deflects zero tickets
8. Ticket-Form Deflection Step	Low (18%)	Low (1 wk)	High	Medium	Frustrates merchants by gating human support access

Strategic Choice: Alternative #1 (In-App Co-Pilot) combined with #2 (Input Linter) was chosen because non-technical merchants do not want documentation—they want their broken campaign fixed immediately with 1 click.

• SYSTEM ARCHITECTURE & COMPLIANCE

Cheerio Co-Pilot: Technical Architecture & Operational Guardrails

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2.1 PRODUCT SPECIFICATION: THE IN-APP CONTEXT-AWARE CO-PILOT

Cheerio Co-Pilot is embedded directly into campaign workflows (WhatsApp Templates, CSV Audiences, Phone Registration). Unlike conversational chatbots that live in a detached widget, Co-Pilot maintains real-time DOM telemetry: form field values, draft history, account verification status, and Meta Cloud API HTTP response headers. When a merchant makes an error, Co-Pilot provides two transparent, 1-click remediation paths on-screen:

Path A: Clean Utility (1-Click Auto-Fix) — Meta Rate: ₹0.1150/msg

Strips violating promotional phrases (e.g., *"HURRY! 50% OFF Diwali Sale"*) while preserving dynamic variables ({{1}}). Rewrites message into strict transactional compliance with live phone diff preview, 1-click rollback, and disclaimer that final approval rests with Meta.

Path B: Reclassify to Marketing — Meta Rate: ₹0.8631/msg

Preserves the merchant's promotional discount copy, automatically reassigns the template category to Marketing, and alerts the merchant to the higher Meta delivery tariff (₹0.8631 vs. ₹0.1150 base rate), preventing surprise billing disputes.

2.2 MULTI-TIER RESOLUTION ENGINE & COST-OPTIMIZED PIPELINE

To deliver fast user responsiveness and keep raw LLM inference costs below ₹0.20 per query (blended fully loaded cost is ₹1.20/query including vector search, hosting, Redis cache, and monthly tooling), Co-Pilot utilizes a 3-tier processing pipeline:

Tier 1: Client Regex Linter — Local • Free

Deterministic browser checks evaluate variable counts, whitespace, and promo keywords. Traps 100% of missing country codes on CSV uploads without making an LLM API call.

Tier 2: Semantic Vector Cache — Edge • Low Cost

An Upstash Redis semantic cache matches inbound copy against verified compliant template patterns. Returns instant cached rephrasings for standard Indian festive campaigns.

Tier 3: GPT-4o-mini LLM — Cloud API • ₹0.18

Invoked only for novel, complex copy rephrasing or Hinglish intent parsing. Uses structured JSON outputs with prompt caching (50% input discount) to deliver exact replacement payloads.

2.3 VERIFIED META CLOUD API ERROR TAXONOMIES & INDIA RATE CARDS

META ERROR / REASON	OFFICIAL META SPECIFICATION	VERIFIED RATE / REFERENCE	CO-PILOT AUTONOMOUS ACTION
INCORRECT_CATEGORY	Template submitted as Utility contains promotional phrases or commercial CTAs.	India Utility: **₹0.1150/msg** India Marketing: **₹0.8631/msg** *developers.facebook.com*	Pre-flight linter intercepts template prior to submission; offers 1-click copy strip or category shift.
Code 133010 ACCOUNT_NOT_REGISTERED	Phone Number ID not registered on Cloud API; coexistence lock with mobile WhatsApp app.	Meta Registration Endpoint: `POST /{phone-id}/register` Requires 6-digit PIN	Guides merchant through 2-step verification PIN registration; triggers cloud API handshake.
Code 132015 TEMPLATE_PAUSED	Template paused by Meta algorithms due to low recipient quality rating or spam reports.	3 to 6 hour cooling-off period before permanent disablement (Code 132016).	Informs merchant of cooling period; enforces audience segment exclusion for opt-outs.

2.4 2-FAILURE ESCALATION CIRCUIT & STRUCTURED DIAGNOSTIC TELEMETRY PACKET

If a merchant struggles or the automated fix fails **twice consecutively**, Co-Pilot trips a deterministic circuit breaker. Instead of trapping the user in an infinite bot loop, Co-Pilot compiles a structured JSON Diagnostic Packet directly to the human support queue:


```
{ "packet_uuid": "PKT-2026-CHRIO-8921", "merchant_id": "FABJEWELS_D2C", "waba_id": "99182310892", "phone_id": "109923812903",
"meta_error_code": 133010, "meta_error_subcode": "ACCOUNT_NOT_REGISTERED", "failed_attempts": 2, "circuit_breaker_tripped": true,
"client_state": { "target_phone": "+919876543210", "pin_submitted": true, "mobile_deregistration_pending": true },
"suggested_human_action": "Initiate Meta backend provisioning reset to clear SMB coexistence lock on phone_id 109923812903." }
```


2.5 OPERATIONAL GUARDRAILS & EXPLICIT REFUSAL BOUNDARIES

Financial & Invoicing Disputes — Hard Refusal

Policy: Co-Pilot is strictly prohibited from executing financial adjustments or issuing refund guarantees. Any query regarding credit card overcharges, failed renewals, or invoice adjustments routes instantly to the Human Finance Desk with account billing telemetry.

Meta Ban & License Boundaries — Zero-Override

Policy: Cheerio cannot arbitrarily override Meta's compliance servers. When accounts are flagged for spam, Co-Pilot refuses to promise an unban. It generates the official appeal dossier and assigns it directly to Cheerio's Meta Compliance Lead.

• STRATEGY, PACKAGING & GOVERNANCE

Metrics Framework, Monetization Strategy & Audit Protocols

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3.1 NORTH STAR METRIC & REALISTIC 90-DAY TRAJECTORY (NEW ONBOARDINGS)

Cheerio Co-Pilot optimizes for a single customer success metric: **Zero-Touch Onboarding Completion Rate (ZOCR)**—the percentage of newly registered merchants who successfully connect their WABA, verify an audience, and schedule their first campaign without raising a human support ticket. Assuming an onboarding cohort of **~500 new merchant signups/month**, an 11.0% lift delivers **55 additional merchants** completing onboarding autonomously each month (660/year):

22.0%	33.0%	55 / mo	< 5.0%
BASELINE ZOCR [Candidate Estimate]	DAY 90 TARGET +11.0% Realistic Gain	ADDED ACTIVE BIZ From 500 Signups Cohort	7-DAY REOPEN RATE Quality Guardrail

CO-PILOT PRODUCT INTERVENTION	TARGET ZOCR LIFT	UNDERLYING OPERATIONAL MECHANISM & VERIFICATION
Template Pre-Flight Linter	+5.5%	Eliminates 1st-attempt template rejections via real-time regex & category correction.
CSV Audience Sanitizer	+3.0%	Fixes missing +91 dial codes and spacing errors in browser before broadcast launch.
WABA Phone Registration Diagnostic	+2.5%	Guides 2-step verification PIN setup; unblocks SMB mobile coexistence traps.
Total 90-Day Achievable Expansion	**+11.0% (22% → 33%)**	**Directly unblocks 55 additional newly registered merchants per monthly cohort.**

3.2 STRICT DEFLECTION DEFINITION & CUSTOMER QUALITY GUARDRAILS

Audit-Proof Deflection Standard: A support interaction is counted as "Deflected" *only* if the merchant completes their workflow in-app and **raises no human ticket or reopened inquiry within 7 calendar days**. Users who abandon the workflow in frustration are explicitly categorized as "Failed Deflections" to prevent false reporting.

Guardrail 1: Post-Remediation CSAT — Target: ≥ 4.5 / 5.0

1-click rating displayed immediately after auto-fix. If a specific linter rule falls below 4.0 CSAT, it is demoted to "Human Review Only" automatically.

Guardrail 2: Escalation Telemetry — Target: ≥ 95%

Monitors percentage of human-escalated tickets containing a valid Diagnostic Packet. Ensures support reps never ask merchants for account ID or screenshots.

4.1 MONETIZATION & BILLING STRATEGY OPTIONS (LINKED MODEL FORMULAS)

PRICING MODEL	PRICE	FULLY LOADED COST	MARGIN (%)	RECOMMENDATION & VALUE RATIONALE
1. Free Core Co-Pilot (Recommended Base)	₹0 / mo	₹1.20 / query (LLM+Cache+Tooling)	Subsidized	**Recommended Base:** Generates ₹1.58L/mo net surplus (Net Retention MRR ₹1.76L minus AI cost ₹18.5k).
2. Priority Support Add-on (Recommended Upsell)	**₹499 / mo**	**₹21.00 / merchant (5 queries + SLA)**	**+95.8%**	**Recommended Upsell: Dedicated WhatsApp agent, priority SLA <15 mins. Formula: (499-21)/499 = 95.8% gross margin.**
3. Per-Resolution Micro-Fee	₹5.00 / fix	₹1.20 / query	+76.0%	Creates onboarding barrier; disincentivizes self-service; provokes SMB backlash.
4. Prepaid AI Resolution Pack	₹199 / 50 fixes	₹60.00 / pack	+69.8%	High gross margin, but adds wallet friction during the critical first campaign setup.

[Fully loaded AI query cost (₹1.20) = LLM token (₹0.20) + Redis/edge cache (₹0.14) + Monthly tooling per query (₹0.86). Amortized human escalation share (₹14.00) is shown as a separate line].

4.2 OBSERVABILITY, SAFETY THRESHOLDS & MONTHLY AUDIT PROTOCOL

AUDIT PARAMETER / CHECK	ALERT THRESHOLD	SEVERITY	ASSIGNED OWNER	REMEDIATION ACTION
Cost per Resolved Query	> 2× Baseline (> ₹0.40)	High	Lead APM / ML Eng	Audit prompt token usage; revert to client regex linter.
Session Token Expansion	> 140% of 7-day avg	Medium	Backend Lead	Prune conversation history; compress error JSON payloads.
Merchant Session Spikes	> 50 sessions / 24 hrs	High	SecOps / CS Lead	Rate-limit merchant; initiate proactive account audit.
Deflection Rate (7-day verified)	< 40% (Target: 65%)	Critical	Product Manager	Audit Meta API error changes; update linter taxonomies.
Zero-Touch Onboarding (ZOCR)	< 25% (Target: 33%)	High	Onboarding Lead	Review Day 1-7 drop-off funnels; audit WABA binding failures.
7-Day Ticket Reopen Rate	> 5.0% of deflections	High	QA / Support Lead	Investigate auto-fix false positives; refine regex patterns.
Invoice Reconciliation	OpenAI bill vs logs > ±2%	Medium	Finance / Tech Lead	Reconcile dropped webhooks, retries, or orphaned sessions.

• FINANCIAL MODEL & ENGINEERING LOG

Financial Defensibility, Retrospective & Deliverables

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5.1 FINANCIAL MODEL & AVOIDED HEADCOUNT ECONOMICS (5,000 MERCHANTS)

FINANCIAL PARAMETER	PURE HUMAN MODEL	WITH CHEERIO CO-PILOT	NET DELTA / IMPACT	MODEL REFERENCE / FORMULA
Inbound Query Volume	17,500 queries / mo	17,500 queries / mo	0 (Constant)	Volume_Model!B7 (5,000 × 3.5)
Autonomous Deflection	12.0% (FAQ baseline)	65.0% (Co-Pilot target)	+53.0% Deflected	11,375 queries resolved by AI
Human Headcount Required	20 Support Agents	7 Support Agents	**-13 Avoided Hires**	Avoided hiring as volume expands
Monthly Human Payroll (₹)	₹7,00,000 / mo	₹2,45,000 / mo	-₹4,55,000 / mo	Human_Comparison!B12 vs B18 (@ ₹35k)
Monthly AI & Tooling Cost (₹)	₹0 / mo	₹18,500 / mo	+₹18,500 / mo	LLM tokens (₹3,500) + Tooling (₹15,000)
Net Monthly Support Cost (₹)	**₹7,00,000 / mo**	**₹2,63,500 / mo**	**-₹4,36,500 / mo**	**₹52.38 Lakhs / year savings**
Cost per Resolved Ticket (₹)	**₹40.00 / ticket**	**₹15.06 / ticket**	**-₹24.94 (-62.4%)**	**Human_Comparison!B13 vs B22**
CapEx Dev Build & Payback	-	₹2,50,000 build cost	**0.57 Months Payback**	~17 days payback period on build

5.2 HYPOTHETICAL RETENTION SCENARIO & DUAL-BASE ROI (NO DOUBLE COUNTING)

To maintain mathematical integrity, support operational savings and retention revenue are modeled separately as an illustrative 90-day pilot cohort scenario (not an empirical forecast). With 5,000 active businesses paying ₹2,999/mo (₹1.50 Cr MRR) experiencing 5.0% baseline churn:

60 Merchants
RETAINED / MONTH
1.2% Pilot Target Drop

₹1,61,440
NET PRESERVED MRR
Gross ₹1.80L less ₹18.5k Total AI

8.7× Net ROI
ON TOTAL PROGRAM SPEND
50.4× Net ROI on Direct Tokens

MODEL SCENARIO	CHURN REDUCTION	PRESERVED MRR (GROSS / NET)	NET ROI (TOKENS VS. TOTAL PROGRAM)
Conservative (Low)	0.5% absolute	₹74,975 / ₹56,475	**20.4×** (Tokens) • **3.1×** (Total Spend)
Expected (Base)	**1.2% absolute**	**₹1,79,940 / ₹1,61,440**	**50.4× (Tokens) • 8.7× (Total Spend)**
Aggressive (High)	2.0% absolute	₹2,99,900 / ₹2,81,400	**84.7×** (Tokens) • **15.2×** (Total Spend)

6.1 AI COLLABORATION & ENGINEERING RETROSPECTIVE (PART 3)

AI Tools Used: Claude 3.5 Sonnet / Antigravity Agent (architecture, Meta API error taxonomy, financial modeling); Cursor & Tailwind CSS (interactive prototype development); ChatGPT-4o (support benchmark validation).

3 Key Prompts That Worked:

1. *Meta Error Taxonomy:* "Break down top failure codes for WhatsApp templates on Meta Cloud API (INCORRECT_CATEGORY, 133010, 132015) with programmatic actions."
2. *Financial Token Economics:* "Calculate token costs for 17,500 queries on GPT-4o-mini ($0.15 in, $0.60 out, $0.075 cache) vs. 40 tickets/day Indian support salary (₹35k/mo)."
3. *Prototype Architecture:* "Create interactive Material 3 UI with 4 scenarios: pre-flight linter, 2-failure circuit breaker, refusal guardrail, and phone binding 133010."

Failed Prompt & Iteration: Initial query asked to "calculate ROI replacing support agents". AI assumed 100 tickets/day (2.5× industry capacity) and divided gross revenue by cost. Corrected prompt to 40 tickets/day, 22 workdays (880 tickets/mo), avoided hiring framing, and explicit gross/net ROI separation.

AI Hallucination Caught: AI suggested overriding Meta's backend approval status. Caught: Cheerio as an official BSP cannot override Meta compliance servers. Corrected to a client-side pre-flight linter that rephrases copy before API submission.

Candidate Plain-Voice Reflection: "I chose Assignment A because onboarding support friction directly caps merchant activation—if an SMB's first WhatsApp broadcast is rejected by Meta, they churn before month two. I initially doubted whether non-technical shop owners would trust an automated AI rewrite over a human agent, which is why I designed 1-click diff previews with explicit tariff warnings rather than silent background auto-fixes. The biggest decision I changed my mind on was pricing: I initially considered a per-resolution fee, but modeling the economics showed that retaining even 1.2% more merchants via free self-serve fixes yields ₹1.61L/month in net preserved MRR—making a free core copilot far more valuable to Cheerio's bottom line than charging users for support."

6.2 VERIFIED ASSIGNMENT DELIVERABLES & ARTIFACT ACCESS

DELIVERABLE ARTIFACT	DESCRIPTION & TECHNICAL SCOPE	ACCESS LINK / VERIFICATION
1. Interactive Web Prototype	4 Scenarios: Pre-Flight Linter, Circuit Breaker, Refusal, WABA Phone Binding (Code 133010) + Edge cases.	saksham-gupta-gh.github.io/cheerio-copilot/ ✓ Live on GitHub Pages (Mobile Ready)
2. Financial Model (Google Sheet)	8-tab formulaic model: Assumptions, Volume, Tokens, Headcount, Churn ROI, Billing, and Auditing.	Google Sheets Live Formula Model ✓ Live Formula Model (View Access Enabled)
3. Engineering Prompt Log	Complete transparent engineering log of prompts, token costs, hallucinations caught, and iterations.	prompt_log.md (GitHub Source) ✓ Full Transcripts & Error Taxonomy Audited